# A Fair and Efficient Way of Cost Sharing in Collaborative VRP

Ugur Guler[1,2], Andre Berger[1], Thijs Veugen[2,3] , Frank Phillipson[1,2]
[1] Quantitative Economics, School of Business and Economics, Maastricht University, Maastricht, The Netherlands
[2] TNO, The Hague, The Netherlands
[3] Faculty of Electrical Engineering, Mathematics and Computer Science, University of Twente, Enschede, The Netherlands

**Abstract**

## 1 Introduction

The Vehicle Routing Problem (VRP) is a cornerstone of operations research and logistics optimization, with applications ranging from freight transportation to last-mile delivery and supply chain management. The classical VRP considers a single decision-maker who seeks to design cost-efficient vehicle routes while satisfying capacity and demand constraints. Over the past decades, numerous exact and heuristic approaches have been proposed to tackle VRP and its many extensions, such as time windows, heterogeneous fleets, and multi depot settings.

In modern logistics, however, competition and fragmentation often lead to inefficiencies, as companies plan their routes independently. To address this, the concept of *Collaborative Vehicle Routing* has gained increasing attention (Cruijssen, 2020). In collaborative settings, companies form coalitions, share depots and customers, and coordinate their routes with the aim of reducing total costs. Empirical evidence and case studies show that collaboration can yield substantial savings, improve resource utilization, and reduce environmental impact through fewer vehicle kilometers traveled (Cruijssen et al., 2007b).

The need for collaboration has become even more critical with the rapid expansion of e-commerce and the growing pressure on logistics providers to deliver faster and more sustainably. Increasing customer expectations and competitive market conditions force smaller carriers to manage their resources more efficiently. Collaboration allows them to exchange complementary orders, jointly plan their operations, and benefit from economies of scale. Beyond economic efficiency, cooperation also supports environmental goals. According to the European Environment Agency (EEA), road transport accounts for the largest share of greenhouse gas (GHG) emissions in the transport sector, representing about 73% of the EU's total transport emissions in 2022 (EEAwebsite).

Collaborative logistics can thus contribute to reducing both operational costs and carbon emissions.

Collaboration can take various forms. In *horizontal collaboration*, companies operating at the same level of the supply chain, such as two parcel delivery firms, share routes, depots, or vehicles to avoid redundant trips. In *vertical collaboration*, actors from different levels of the supply chain, such as a manufacturer and a distributor, coordinate their routing and inventory decisions to improve overall efficiency. Both approaches offer clear advantages in terms of cost, emissions, and service quality but also require trust and information sharing between partners (Gansterer and Hartl, 2018). Moreover, collaborations can be organized in different decision structures. In *centralized collaboration*, a central planner computes the joint routing plan with full information. In *decentralized collaboration*, companies maintain autonomy and reach agreements through negotiation or order exchange processes (Gansterer and Hartl, 2018).

While collaboration yields global efficiency gains, it also raises a key challenge: how to fairly distribute the resulting savings among the participating companies. Without a fair and transparent allocation rule, coalitions may become unstable and cooperation unsustainable. Cooperative game theory provides a natural framework to tackle this issue by modeling the collaboration as a game with transferable utility. Classical solution concepts such as the *Shapley value*, *core*, *nucleolus*, and *proportional rules* have been widely applied to allocate costs in collaborative logistics (Guajardo and Rönnqvist, 2016). These concepts ensure fairness and stability in theory, but they require evaluating the *characteristic function* (the minimum routing cost for every possible coalition of companies) which is computationally intractable for large problems. Since each coalition evaluation involves solving a VRP, whose complexity grows exponentially with the number of companies, these methods become impractical in real world applications.

This paper addresses this gap by developing a unified heuristic algorithm that simultaneously optimizes routes and allocates costs in collaborative logistics. We focus on centralized horizontal cooperation, where companies jointly plan routes through a shared depot. We keep the problem setup and proposed solution method intentionally simple to demonstrate that our approach is intuitive and easily adaptable to other VRP variants and solution techniques. The method first constructs initial routes for each company independently using the Clarke and Wright savings algorithm, and then merges these routes into a collective plan. A deterministic local search procedure iteratively improves the joint routes by reallocating customers across companies. Each time the algorithm identifies a cost saving move, the savings are distributed among the affected companies using a proportional cost sharing rule. This dynamic process continuously refines both the routing plan and the cost allocation vector, ensuring fairness and computational efficiency without the need to compute the full characteristic function.

The main contribution of this study is the development of a computationally efficient framework that integrates route construction, cooperative improvement, and fair cost allocation in a single process. The approach maintains transparency and scalability while providing a theoretically grounded allocation mechanism. We show that the proposed algorithm satisfies three fundamental fairness axioms such as *efficiency*,

*individual rationality*, and *dummy player* and preserves symmetry under certain idealized conditions. The algorithm is tested on both small illustrative examples and benchmark VRP instances, where its results are compared with classical allocation concepts such as the Shapley value, the nucleolus, and the egalitarian rule. The findings demonstrate that the proposed method achieves a strong balance between fairness and computational tractability, offering a viable alternative to exact cooperative game theoretic methods in collaborative routing.

The remainder of this paper is structured as follows: Section 2 reviews the existing literature on collaborative VRP, particularly horizontal collaboration and cost allocation methods, and categorizes various methodologies related to the subject. Solution concepts from cooperative game theory, the mathematical model of Capacited Vehicle Routing Problem (CVRP) and its extension to collaborative setup are briefly discussed in Section 3. Section 4 presents our proposed deterministic local search algorithm and its pseudocode. In Section 5, theoretical results of the heuristic algorithm on the cooperative game theory and numerical tests on small instances are discussed. In Section 6, we present the results of the algorithm on a real world application. Finally, in Section 8 we conclude our study and discuss potential future research directions.

# 2 Related Work

The Vehicle Routing Problem (VRP) was first introduced by Dantzig and Ramser (1959) as the *truck dispatching problem*, aiming to design cost-minimizing routes for gasoline delivery. Since then, VRP has become one of the most extensively studied combinatorial optimization problems in operations research. Numerous variants have been developed to reflect practical requirements, including the Capacitated VRP (CVRP), the VRP with Time Windows (VRPTW), and the Multi-Depot VRP (MDVRP). A detailed survey of these variants can be found in Mor and Speranza (2022). Due to its NP-hard nature, exact methods such as branch and bound and branch and cut are effective only for small instances. For larger problem sizes, heuristic and metaheuristic approaches such as the Clarke and Wright savings algorithm, tabu search, simulated annealing, genetic algorithms, and variable neighborhood search remain the primary tools for solving VRPs in practice (Archetti and Speranza, 2014).

While traditional VRP assumes a single decision maker, real-world logistics often involves multiple companies. The shift to *Collaborative Vehicle Routing*, where companies jointly plan routes, has been shown to generate significant efficiency gains. Empirical findings indicate that horizontal collaboration can reduce total transportation costs by 5% to 30% (Cruijssen et al., 2007b; Cruijssen and Salomon, 2004), with some studies reporting savings of up to 46% (Guajardo and Rönnqvist, 2016). For instance, a case study on a local courier service demonstrated that cooperative route planning outperformed non-cooperative operations by up to 20% (Lin, 2008). These benefits are achieved through reduced empty mileage, joint routing, and improved vehicle utilization. Beyond economic gains, collaboration also supports environmental goals by lowering

fuel consumption and emissions (Ballot and Fontane, 2010).

Despite these clear advantages, the practical adoption of collaborative routing remains limited. A key impediment is the challenge of designing mechanisms that allocate shared costs and benefits fairly among participants (Cruijssen et al., 2007b). Gansterer and Hartl (2018), in their comprehensive review of collaborative VRPs, highlight this fair cost allocation as a major implementation challenge.

*Cooperative game theory* provides a natural framework for addressing the allocation problem in collaborative routing. Allocation mechanisms such as the *core*, the *Shapley value* (Shapley, 1953), the *nucleolus* (Schmeidler, 1969), and *proportional rules* (for instance, the egalitarian method introduced by Dror and Trudeau (1990)) distribute costs according to fairness and stability principles. Applications in logistics and transportation confirm the suitability of these mechanisms (Engevall et al., 2004; Özener and Ergun, 2008; Frisk et al., 2010; Flisberg et al., 2015). However, these approaches share a critical limitation: computing the characteristic function requires solving a VRP for every possible coalition. As the number of coalitions grows exponentially with the number of companies, the approach becomes computationally infeasible beyond small scale instances. This scalability barrier has become one of the main obstacles to implementing cooperative routing mechanisms in real-world logistics (Guajardo and Rönnqvist, 2016).

To overcome this computational bottleneck, several research directions have emerged. Sampling based methods approximate the Shapley value using Monte Carlo or polynomial time sampling (Castro et al., 2009; Mitchell et al., 2022), while other studies propose proxy functions that estimate coalition values through surrogate distance or savings-based measures (Guajardo et al., 2018). Although these approaches reduce the number of required VRP evaluations, they typically compromise fairness guarantees or yield unstable allocations.

Other related works explore fairness from alternative perspectives. López Sánchez et al. (2024) analyze cooperative fleets under egalitarian and elitist social welfare objectives, highlighting trade-offs between efficiency and fairness. Lozano et al. (2013) evaluate various cost allocation methods by examining the Mean Absolute Deviation in the distribution of cost savings among different companies. Audy et al. (2012) study the optimization of inter-firm collaboration processes and introduce five coordination mechanisms aimed at facilitating information sharing, logistical activity coordination, and profit distribution. Optimization based frameworks have also been developed to incorporate fairness directly into the routing problem. Ohmori and Yoshimoto (2021), for instance, formulate a max-min fairness model to ensure equitable benefit distribution, while van Zon et al. (2021) propose a row generation algorithm to compute core allocations efficiently. Despite these advances, most of these approaches treat routing and allocation as two separate stages: first optimizing routes and then distributing costs according to fairness rules. Moreover, the fairness criteria investigated in these works often lack strong theoretical foundations or are only applicable in specific contexts.

Recent studies have attempted to integrate learning based or heuristic strategies. Multi agent reinforcement learning (MARL) frameworks, for example, treat companies as agents negotiating cooperative solutions without explicitly computing the characteristic

function (Mak et al., 2023). While promising in avoiding the explicit computation of the characteristic function, such models require extensive training and often fail to guarantee fairness axioms. Moreover, learning based approaches can struggle to explain high-level decisions that would be meaningful for decision makers (Gunning, 2017). Similarly, heuristic based hybrid models (Shi et al., 2020) improve scalability but still separate the optimization and allocation phases, limiting their fairness and efficiency consistency.

The literature reveals a persistent gap between *computational efficiency* and *fairness assurance* in collaborative routing. Cooperative game-theoretic methods ensure strong fairness properties but are computationally intractable for large-scale collaborations. In contrast, heuristic or learning based approaches are often fail to guarantee stability or fairness principles. Furthermore, most existing models decouple routing and cost allocation, solving them sequentially rather than as a unified decision process.

# 3 Problem Definition

This section formally defines the collaborative vehicle routing problem within the framework of cooperative game theory. We first introduce the key cooperative game theoretic concepts that support our analysis of cost allocation. Subsequently, we present the underlying mathematical model, the Capacitated Vehicle Routing Problem (CVRP), which is used to compute the cost savings exactly for any potential coalition of companies. The combination of these two frameworks allows us to model both the collaborative logistics problem and the economic problem of fair profit sharing.

## 3.1 Cooperative Game Theory Background

In the cooperative setting, companies can be modeled as players who may form coalitions in order to reduce costs through joint operations. Cooperative game theory provides the formal tools to study such situations, evaluate the benefits of cooperation, and design fair allocation rules for distributing the coalition gains among participants.

Formally, a *cooperative game* is defined by a finite set of players $M = \{1, 2, \ldots, m\}$, referred to as the *grand coalition*, together with a *characteristic function* $\nu : 2^{|M|} \to \mathbb{R}$ satisfying $\nu(\emptyset) = 0$. For each coalition $C \subseteq M$, the value $\nu(C)$ represents the total cost that coalition $C$ can achieve when its members cooperate. A *coalition structure* is a partition of $M$ into disjoint subsets, each subset being a coalition.

A central object of interest is the *cost allocation vector* $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_m) \in \mathbb{R}^m$, where $\phi_i$ denotes the cost allocated to player $i \in M$. An allocation is feasible if it distributes the total cost without exceeding the value of any coalition, i.e., $\sum_{i \in C} \phi_i \leq \nu(C)$ for every coalition $C \subseteq M$, with $\phi_i \geq 0$ for all $i$. A *solution concept* is a rule that assigns at least one such cost vector to each cooperative game. The challenge is to allocate the total cost $\nu(M)$ among the players in a fair and stable manner.

Various solution concepts have been proposed, each based on a different notion of

fairness. These concepts are typically evaluated against a set of desirable properties. Important properties include: **existence**, which ensures a solution can be found for a broad class of games; **uniqueness**, which guarantees a single allocation for each game; **efficiency**, meaning the entire cost is allocated; and **symmetry**, which requires that identical players receive the same allocation. Other key properties are **rationality**, ensuring no player or coalition has an incentive to deviate; the **dummy player property**, where a player who contributes nothing to any coalition bears no cost; and **linearity**, which ensures the solution is consistent under linear transformations of the game.

One of the most prominent solution concepts is the *Shapley value*, which allocates to each player its expected marginal contribution across all possible coalition formation orders:

$$\phi_i(\nu) = \sum_{C \subseteq M \setminus \{i\}} \frac{|C|!(m - |S| - 1)!}{m!} \big[\nu(C \cup \{i\}) - \nu(C)\big], \quad i \in m. \tag{1}$$

The Shapley value is the unique solution concept that satisfies the four axioms of efficiency, symmetry, the dummy player property, and linearity. For these reasons, it is widely regarded as a fair and principled allocation rule.

Another fundamental solution concept is the *core*, defined as the set of allocations such that no coalition can improve upon them:

$$\sum_{i \in C} \phi_i \geq \nu(C) \quad \text{for all } C \subseteq M.$$

The core captures the notion of stability, since no subset of players has an incentive to deviate from the proposed allocation (rationality). However, the core may be empty in many games. When it is non-empty, another solution concept the *nucleolus* provides a unique allocation within the core. It is defined by minimizing, in a lexicographic sense, the excesses of coalitions, where the excess of $C \subseteq M$ under allocation $\boldsymbol{\phi}$ is given by

$$e(C, \boldsymbol{\phi}) = \nu(C) - \sum_{i \in C} \phi_i.$$

The nucleolus equalizes coalition dissatisfaction as much as possible, offering a compelling balance between fairness and stability. Its main drawback is computational: finding the nucleolus requires complete characteristic function and then solving a sequence of linear programs.

Besides these theoretically grounded concepts, proportional rules are often employed in practice. Under such rules, the value of the grand coalition $\nu(M)$ is divided among players according to exogenous weights, such as demand shares, fleet sizes, or historical contributions. One of the proportional rules called *Equal Surplus Division (ESD)* allocates the surplus generated by a coalition of players equally. Although these rules lack strong axiomatic justification, they are simple, transparent, and computationally efficient, which makes them attractive in real-world applications.

However, it is important to note that most principled solution concepts are computationally demanding. The Shapley value, the core and the nucleolus requires evaluating the complete characteristic function across all $2^{|M|}$ coalitions. Moreover, the nucleolus involves repeatedly solving linear programs. Such requirements quickly become impractical in large-scale combinatorial optimization problems, such as those arising in vehicle routing. In the cooperative CVRP formulation that follows, this compound complexity arising from both the optimization problem and the allocation mechanism will become particularly apparent.

## 3.2 Mathematical Model of CVRP and Its Extension to Cooperative VRP

Cooperation among companies can occur in various forms, such as sharing customers, vehicles, depots, or other service facilities. In this study, we focus on cooperation through customer sharing, where companies jointly serve their combined set of customers using a central depot and a common fleet of vehicles. This simplification allows us to concentrate on the methodological aspects of our approach rather than on modeling additional operational complexities. Nevertheless, the proposed methodology can be readily adapted to other variants of the Vehicle Routing Problem (VRP) or to different levels of cooperation.

Let $M = \{1, \ldots, m\}$ denote the set of companies, where each company $m \in M$ serves its own set of customers denoted by $C^m$. When a subset of companies decides to cooperate, they form a coalition $S \subseteq M$. The cooperating companies pool their customers into a joint set

$$C(S) = \bigcup_{m \in S} C^m,$$

which is served collectively from a shared central depot $d_0$. A homogeneous fleet of vehicles, each with capacity $Q$, is available for deliveries. The travel cost between any two locations is represented by the distance matrix $D = (d_{ij})$. The cooperative planning problem is then modeled as a *Capacitated Vehicle Routing Problem (CVRP)*, where the customer set is determined by the coalition $S$ instead of a single company.

**Model Parameters and Sets**

| | |
|---|---|
| $M$ | Set of all companies, $M = \{1, \ldots, m\}$. |
| $S$ | Subset of companies forming a coalition, $S \subseteq M$. |
| $C^m$ | Set of customers belonging to company $m \in M$. |
| $C(S)$ | Combined customer set for coalition $S$. |
| $K$ | Set of available vehicles, $K = \{1, \ldots, k\}$. |
| $d_0$ | Central depot location. |
| $Q$ | Vehicle capacity ($Q > 0$). |
| $q_i$ | Demand of customer $i \in C(S)$, where $0 < q_i \leq Q$. |
| $D$ | Distance (or cost) matrix, where $d_{ij}$ is the travel cost between $i$ and $j$. |

**Decision Variables**

$$x_{ij} = \begin{cases} 1 & \text{if a vehicle travels directly from node } i \text{ to node } j, \\ 0 & \text{otherwise,} \end{cases} \qquad \forall i, j \in \{d_0\} \cup C(S),\ i \neq j,$$

$u_i =$ Cumulative load on a vehicle after visiting customer $i \in C(S)$.

**Objective Function**

The goal is to minimize the total travel cost of all vehicles serving the coalition $S$:

$$\text{minimize} \quad \sum_{i \in \{d_0\} \cup C(S)} \sum_{\substack{j \in \{d_0\} \cup C(S) \\ j \neq i}} d_{ij} \cdot x_{ij}. \tag{2}$$

**Constraints**

$$\sum_{\substack{j \in \{d_0\} \cup C(S) \\ j \neq i}} x_{ij} = 1, \qquad \forall i \in C(S), \tag{3a}$$

$$\sum_{\substack{j \in \{d_0\} \cup C(S) \\ j \neq i}} x_{ji} = 1, \qquad \forall i \in C(S), \tag{3b}$$

$$\sum_{j \in C(S)} x_{d_0 j} \leq |K|, \tag{3c}$$

$$\sum_{i \in C(S)} x_{i d_0} \leq |K|, \tag{3d}$$

$$u_i - u_j + Q \cdot x_{ij} \leq Q - q_j, \qquad \forall i, j \in C(S),\ i \neq j, \tag{3e}$$

$$q_i \leq u_i \leq Q, \qquad \forall i \in C(S), \tag{3f}$$

$$x_{ij} \in \{0, 1\}, \qquad \forall i, j \in \{d_0\} \cup C(S),\ i \neq j. \tag{3g}$$

Constraints (3a) and (3b) ensure that each customer in the combined customer set $C(S)$ is visited exactly once by a vehicle: one incoming arc and one outgoing arc are required at every customer node. Constraints (3c) and (3d) guarantee that no more than $|K|$ vehicles depart from or return to the central depot $d_0$, thus restricting the number of active routes to the available fleet size.

A critical aspect of the formulation lies in the subtour elimination and capacity enforcement, expressed through constraints (3e) and (3f). These are the well-known Miller–Tucker–Zemlin (MTZ) constraints (Miller et al., 1960), which introduce auxiliary continuous variables $u_i$ representing the cumulative load carried by a vehicle upon visiting customer $i$. Constraint (3e) prevents the formation of subtours by enforcing a strict ordering on the customer visits: if a vehicle travels directly from $i$ to $j$, then the load variable at $j$ must be at least the load at $i$ plus the demand $q_j$, unless $j$ is visited by a different vehicle. This logical progression rules out disconnected cycles that do not include the depot. In addition, these constraints simultaneously enforce the vehicle capacity limit $Q$, ensuring that the cumulative demand assigned to any route does not exceed the maximum load a vehicle can carry. Constraint (3f) explicitly bounds the auxiliary variables $u_i$ between the demand $q_i$ and the vehicle capacity $Q$, thereby maintaining consistency with the interpretation of $u_i$ as load levels. Finally, constraint (3g) specifies the binary domain.

The cooperative VRP problem we consider is essentially the classical CVRP, with the distinction that multiple companies to jointly serve their combined customer base $C(S)$ through a shared fleet, thereby embedding cooperation between different stakeholders. This not only enlarges the solution space but also introduces the additional challenge of allocating the resulting costs fairly among participants. Since the CVRP is NP-hard,

the cooperative extension naturally inherits this computational complexity, rendering exact approaches impractical for realistically sized instances.

# 4 Methodology

We propose a two–phase heuristic framework to solve the cooperative vehicle routing problem with cost allocation. The approach integrates an initial route construction heuristic with an improvement phase based on local search and a proportional cost sharing rule. The procedure is summarized in Algorithms 1 and **??**, and visualized in Figure 1.

**Algorithm 1** Clarke–Wright Savings Algorithm

**Input:**

- Customer set $C(S) = \{c_1, c_2, \ldots, c_n\}$ with demands $q_i$
- Central depot $d_0$, vehicle capacity $Q$
- Distance matrix $D = (d_{ij})$

**Output:**

- Set of routes $\mathcal{R}(S)$

```
Initialize n routes: [d_0 → c_i → d_0] for each c_i ∈ C(S)
Initialize savings list S ← ∅
for each pair (c_i, c_j) where i ≠ j do
    s_ij ← d_{d_0 i} + d_{d_0 j} − d_ij                    ▷ Compute savings
    Add (c_i, c_j, s_ij) to S
end for
Sort S in descending order of s_ij
for each (c_i, c_j, s_ij) in sorted S do
    Find route R_i ending with c_i and route R_j starting with c_j
    if R_i ≠ R_j and demand(R_i) + demand(R_j) ≤ Q then
        Merge R_i and R_j: R_new = [d_0 → ... → c_i → c_j → ... → d_0]
        Update R(S) ← (R(S) \ {R_i, R_j}) ∪ {R_new}
    end if
end for
return R(S)
```

**Algorithm 2** Cooperative VRP with Heuristic Cost Allocation

**Input:**

- Set of companies $M = \{1, \ldots, m\}$
- Customer sets $C^m$ for each $m \in M$
- Central depot $d_0$, vehicle capacity $Q$
- Distance matrix $D = (d_{ij})$

**Output:**

- Final set of routes $\mathcal{R}(S)$
- Cost allocation vector $\boldsymbol{\phi} \in \mathbb{R}^{|M|}$

**Phase 1: Independent Route Construction**
**for** each company $m \in M$ **do**
    $\mathcal{R}^m \leftarrow \textsc{ClarkeWrightSavings}(C^m, Q, D)$
    $v(\{m\}) \leftarrow \textsc{TotalDistance}(\mathcal{R}^m)$
**end for**
$\mathcal{R}(M) \leftarrow \bigcup_{m \in M} \mathcal{R}^m, \quad \phi_m \leftarrow v(\{m\})$ for all $m \in M$
**Phase 2: Cooperative Improvement (Local Search)**
**for** $t = 1$ to max_iterations **do**
    best_move $\leftarrow$ None, $\quad \Delta^* \leftarrow 0$
    **for** each neighborhood $N \in \{\text{2-opt}, \text{reinsertion}, \text{swap}\}$ **do**
        **for** each feasible move in $N(\mathcal{R}(M))$ **do**
            Compute cost reduction $\Delta$
            **if** $\Delta > \Delta^*$ **then**
                $\Delta^* \leftarrow \Delta, \quad$ best_move $\leftarrow$ current move
            **end if**
        **end for**
    **end for**
    **if** best_move $\neq$ None **then**
        Apply best_move to $\mathcal{R}(M)$
        Let $S \leftarrow$ set of companies affected by the move
        **for** each $m \in S$ **do**
            $\phi_m \leftarrow \phi_m - w_m(S) \cdot \Delta^*$
            $w_m(S) = \dfrac{\text{share of company } m \text{ in } S}{\sum_{j \in S} \text{share}_j}$
        **end for**
    **else**
        **break**
    **end if**
**end for**
**return** $(\mathcal{R}(M), \boldsymbol{\phi})$

## 4.1 Phase 1: Initial Route Construction

In the first phase, routes are constructed independently for each company using the Clarke and Wright savings algorithm. Let $M = \{1, \ldots, |M|\}$ denote the set of companies, where company $m \in M$ is responsible for serving the customer subset $C_m \subseteq C$. For each company $m$, the Clarke and Wright algorithm generates a feasible set of routes $\mathcal{R}_m$ that satisfies the vehicle capacity constraint $Q$.

The number of vehicles required for company $m$ is approximated by $\lceil |C_m|/Q \rceil$ for simplicity. The cost of serving only company $m$'s customers is computed as

$$v(\{m\}) = \text{cost}(\mathcal{R}_m),$$

which represents the stand-alone cost of company $m$. The collection of initial routes across all companies is given by

$$\mathcal{R}^{(0)} = \bigcup_{m \in M} \mathcal{R}_m,$$

and the initial allocation vector is defined as

$$\phi^{(0)} = (v(\{1\}), v(\{2\}), \ldots, v(\{|M|\}))\,.$$

At this stage, each company fully bears the cost of its own routes without any cooperation. We then pool all initial routes and treat them as a single cooperative solution for the grand coalition $M$, effectively solving one CVRP instance over the combined customer set $C(M) = \bigcup_{m \in M} C_m$.

Since the total number of vehicles used independently, $\lceil \sum_{m \in M} |C_m|/Q \rceil$, can exceed the number required for the grand coalition $\lceil |C(M)|/Q \rceil$, we add dummy customers located at the depot $d_0$ to maintain consistency in vehicle assignment. The number of dummy customers added is

$$N_{\text{dummy}} = \left\lceil \sum_{m \in M} \frac{|C_m|}{Q} \right\rceil - \left\lceil \frac{|C(M)|}{Q} \right\rceil.$$

These dummy customers ensure that surplus vehicles remain idle at the depot, thus not affecting routing costs or allocations in the following phase.

## 4.2 Phase 2: Deterministic Local Search and Cost Allocation

In the second phase, the algorithm iteratively improves the pooled solution through a deterministic local search procedure. At each iteration $t$, the algorithm explores a set of neighborhood operators:

$$N \in \{\text{2-opt}, \text{reinsertion}, \text{swap}\}.$$

Specifically, the *2-opt* operator removes two non-adjacent arcs in a route and reconnects the segments in the reverse order; the *reinsertion* operator selects a single customer and

reinserts it at a different position (possibly in another route); and the *swap* operator exchanges the positions of two customers, either within the same route or across different routes.

Each operator modifies a subset of routes in $\mathcal{R}^{(t-1)}$ and generates a candidate solution $\mathcal{R}'$. The improvement (or saving) achieved by a move is given by

$$\Delta^{(t)} = \text{cost}(\mathcal{R}^{(t-1)}) - \text{cost}(\mathcal{R}').$$

The move with the largest positive $\Delta^{(t)}$ is selected as the best move. If $\Delta^{(t)} > 0$, the move is accepted and the routes are updated:

$$\mathcal{R}^{(t)} \leftarrow \mathcal{R}'.$$

**Coalition identification.** When a move involves customers belonging to multiple companies, the set of affected companies is denoted by

$$S^{(t)} \subseteq M.$$

Only companies in $S^{(t)}$ participate in the cost reallocation resulting from the improvement.

**Cost allocation rule.** The total saving $\Delta^{(t)}$ is distributed among companies in $S^{(t)}$ according to proportional weights. Let $\text{share}_m^{(t)}$ denote the contribution of company $m$ to the modified routes at iteration $t$ (e.g., in terms of demand, distance, or number of involved customers). Then, the relative weight of company $m \in S^{(t)}$ is

$$w_m^{(t)} = \frac{\text{share}_m^{(t)}}{\sum_{n \in S^{(t)}} \text{share}_n^{(t)}}.$$

The cost allocation vector is then updated as

$$\phi_m^{(t)} = \begin{cases} \phi_m^{(t-1)} - w_m^{(t)} \cdot \Delta^{(t)}, & \text{if } m \in S^{(t)}, \\ \phi_m^{(t-1)}, & \text{otherwise.} \end{cases}$$

**Termination.** The iterative process continues until no improving move exists ($\Delta^{(t)} \leq 0$) or a maximum number of iterations is reached. The final outputs are the improved route set $\mathcal{R}^{\text{fin}}$ and the final allocation vector $\phi^{\text{fin}}$.

**Methodology Flow.** The complete process is depicted in Figure 1. The Clarke and Wright algorithm provides initial routes and allocations, which are subsequently refined through iterative neighborhood search, coalition identification, and proportional cost redistribution until the termination condition is met.

## 4.3 Phase 1: Initial Route Construction

In the first phase, routes are constructed independently for each company using the Clarke and Wright savings algorithm. Let $M = \{1, \ldots, m\}$ denote the set of companies, where company $m \in M$ is responsible for serving customers $C_m \subseteq C$. For each company $m$, the Clarke and Wright algorithm generates a feasible set of routes $\mathcal{R}_m$ subject to the vehicle capacity $Q$. The number of vehicles required per company is calculated by $\lceil \frac{C_i}{Q} \rceil$ for the sake of simplicity.

The cost of serving only company $m$'s customers is computed as

$$v(\{m\}) = \text{cost}(\mathcal{R}_m),$$

which represents the stand–alone cost of company $m$. The collection of initial routes is then

$$\mathcal{R}^{(0)} = \bigcup_{m \in M} \mathcal{R}_m,$$

and the initial allocation vector is defined as

$$\phi^{(0)} = (v(\{1\}), v(\{2\}), \ldots, v(\{m\}))\,.$$

At this stage, each company fully bears the cost of its own routes without any cooperation. We then pool all routes and pretend them as the solution of one CVRP called grand coalition $M$. At this point, we adjust the number of vehicles, since $\lceil \sum_{m \in M} |C_m|/Q \rceil \geq \lceil |C(M)|/Q \rceil$, i.e., the required number of vehicles for the grand coalition can be smaller than the total number of vehicles determined in the initial solutions, which may result in inefficient routes. To address this, we add dummy customers, without any company label, located at the depot as needed. The required number of dummy customers is given by

$$N_{\text{dummy}} = \left\lceil \sum_{m \in M} \frac{|C_m|}{Q} \right\rceil - \left\lceil \frac{|C(M)|}{Q} \right\rceil.$$

In this way, the excess vehicles will be assigned to these dummy customers at the depot, ensuring that they do not affect the routing cost or the allocation scheme in the following phase.

## 4.4 Phase 2: Deterministic Local Search and Cost Allocation

In the second phase, the algorithm iteratively improves the pooled solutions through a deterministic local search procedure. At each iteration $t$, the algorithm explores a set of neighborhood operators:

$$N \in \{\text{2-opt}, \text{reinsertion}, \text{swap}\}.$$

Specifically, the *2-opt* operator removes two non-adjacent edges in a route and reconnects the segments in the opposite order, the *reinsertion* operator selects a single

customer and reinserts it at a different position within the same or another route, and the *swap* operator exchanges the positions of two customers, either within the same route or across different routes.

Each operator modifies a subset of routes in $\mathcal{R}^{(t-1)}$ and generates a candidate solution $\mathcal{R}'$. The cost improvement (or saving) associated with a move is given by

$$\Delta = \text{cost}(\mathcal{R}^{(t-1)}) - \text{cost}(\mathcal{R}').$$

The move with the largest positive $\Delta$ is selected as the best move. If $\Delta > 0$, the move is applied, and the solution is updated:

$$\mathcal{R}^{(t)} \leftarrow \mathcal{R}'.$$

**Coalition identification.** When a move involves customers belonging to multiple companies, the set of affected companies is denoted by

$$S^{(t)} \subseteq M.$$

Only companies in $S^{(t)}$ participate in the reallocation of costs resulting from the improvement.

**Cost allocation rule.** The saving $\Delta$ is distributed among companies in $S^{(t)}$ according to proportional weights. Let $\text{share}_m^{(t)}$ denote the contribution of company $m$ to the modified routes (e.g., in terms of demand, distance, or the number of customers involved at that iteration). Then the weight of company $m \in S^{(t)}$ is

$$w_m^{(t)} = \frac{\text{share}_m^{(t)}}{\sum_{n \in S} \text{share}_n^{(t)}}.$$

The allocation vector is updated as

$$\phi_m^{(t)} = \begin{cases} \phi_m^{(t-1)} - w_m^{(t)} \cdot \Delta^{(t)}, & \text{if } m \in S^{(t)}, \\ \phi_m^{(t-1)}, & \text{otherwise.} \end{cases}$$

**Termination.** The iterative procedure continues until no improving move exists ($\Delta^{(t)} \leq 0$) or a maximum number of iterations is reached. The final output is the improved set of routes $\mathcal{R}^{\text{fin}}$ and the cost allocation vector $\phi^{\text{fin}}$.

**Methodology Flow.** The overall procedure is illustrated in Figure 1. The Clarke and Wright algorithm provides initial routes and allocations. The local search procedure then evaluates neighborhood moves, computes savings, identifies affected coalitions, and applies the cost sharing rule. The process iteratively produces new routes and updated allocations until the termination rules are satisfied.

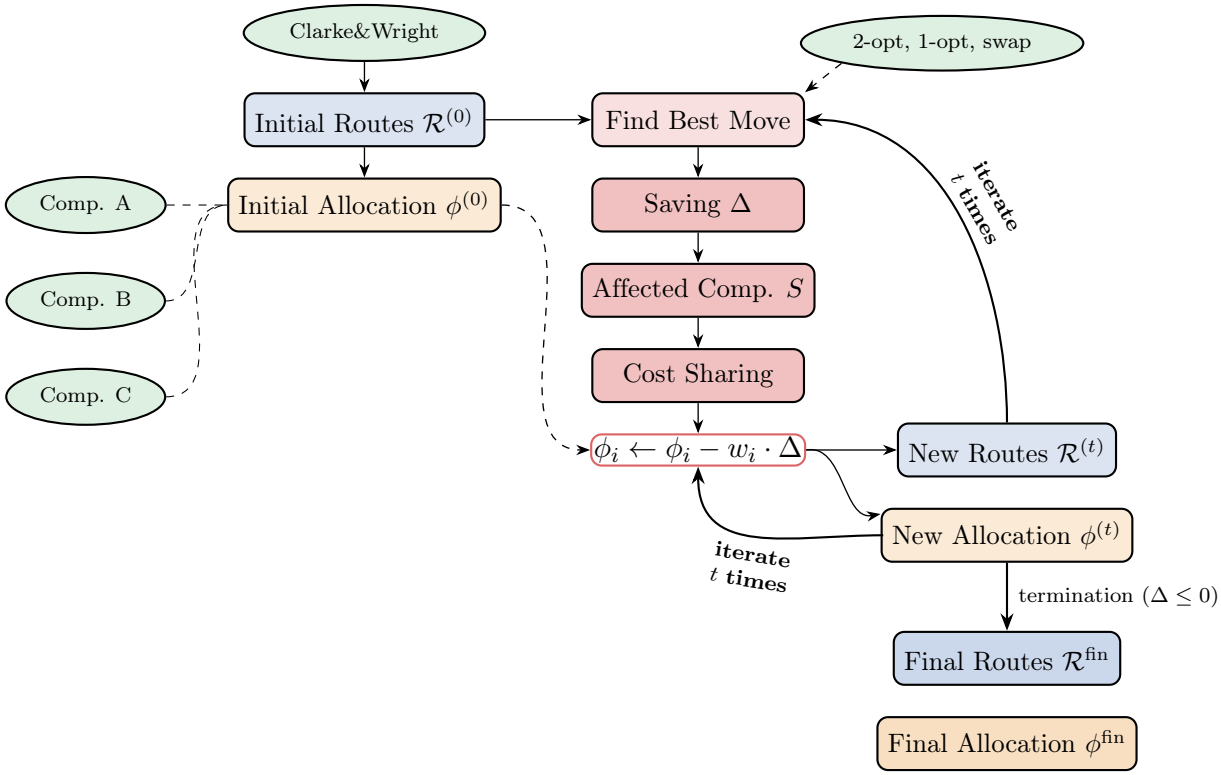


Figure 1: Flowchart of the proposed heuristic algorithm for the cooperative VRP. The process iterates through route improvement and cost allocation updates until no further improving moves can be found. The final output is the improved set of routes $\mathcal{R}^{\text{fin}}$ and the cost allocation vector $\phi^{\text{fin}}$.

# 5 Results

This section presents the results of our study in two complementary parts. First, we provide theoretical results that establish the main properties of the proposed heuristic cost allocation algorithm, including its adherence to standard axioms from cooperative game theory under suitable conditions. These results highlight the conceptual foundations and guarantees of the method. Second, we report computational experiments conducted on randomly generated small instances to assess the algorithm's practical performance. This part focuses on evaluating allocation outcomes relative to established solution concepts, as well as analyzing runtime efficiency and robustness. Together, the two subsections offer both theoretical justification and empirical evidence for the effectiveness of the proposed approach.

## 5.1 Theoretical Results

**Preliminaries.** For each company $i \in M$, let $\mathcal{R}_i^{(0)}$ be the Clarke and Wright routes on $C_i$, and define $v_{\text{alg}}(\{i\}) := \text{cost}(\mathcal{R}_i^{(0)})$ and $\phi_i^{(0)} := v_{\text{alg}}(\{i\})$. Let $\mathcal{R}^{(0)} := \bigcup_{i \in M} \mathcal{R}_i^{(0)}$ as

called pooled initial routes. During the local search, at iteration $t$, the accepted move reduces the cost by $\Delta^{(t)} > 0$ and affects the company set $S^{(t)} \subseteq M$. Weights $w_i^{(t)} \geq 0$ are defined for $i \in S^{(t)}$ with $\sum_{i \in S^{(t)}} w_i^{(t)} = 1$. Allocations are updated as

$$\phi_i^{(t)} \leftarrow \begin{cases} \phi_i^{(t-1)} - w_i^{(t)} \Delta^{(t)}, & i \in S^{(t)}, \\ \phi_i^{(t-1)}, & i \notin S^{(t)}. \end{cases}$$

Let $\mathcal{R}^{\text{fin}}$ and $\phi^{\text{fin}}$ denote the terminal routes and allocation, and set $v_{\text{alg}}(M) := \text{cost}(\mathcal{R}^{\text{fin}})$.

**Theorem 1** (Efficiency)**.** *For all $t \geq 0$,*

$$\sum_{i \in M} \phi_i^{(t)} \;=\; \text{cost}(\mathcal{R}^{(t)}).$$

*In particular,*

$$\sum_{i \in M} \phi_i^{\text{fin}} \;=\; v_{\text{alg}}(M).$$

*Proof.* At $t = 0$, $\text{cost}(\mathcal{R}^{(0)}) = \sum_i \text{cost}(\mathcal{R}_i^{(0)}) = \sum_i v_{\text{alg}}(\{i\}) = \sum_i \phi_i^{(0)}$. At iteration $t$, $\text{cost}(\mathcal{R}^{(t)}) = \text{cost}(\mathcal{R}^{(t-1)}) - \Delta^{(t)}$, while

$$\sum_{i \in M} \phi_i^{(t)} = \sum_{i \in M} \phi_i^{(t-1)} - \sum_{i \in S^{(t)}} w_i^{(t)} \Delta^{(t)} = \sum_{i \in M} \phi_i^{(t-1)} - \Delta^{(t)}.$$

The identity follows by induction on $t$, and at termination yields $\sum_i \phi_i^{\text{fin}} = v_{\text{alg}}(M)$. □

**Theorem 2** (Individual rationality)**.** *For each $i \in M$,*

$$\phi_i^{\text{fin}} \;\leq\; \phi_i^{(0)} \;=\; v_{\text{alg}}(\{i\}).$$

*Proof.* At each iteration, if $i \notin S^{(t)}$ then $\phi_i^{(t)} = \phi_i^{(t-1)}$; if $i \in S^{(t)}$ then $\phi_i^{(t)} = \phi_i^{(t-1)} - w_i^{(t)} \Delta^{(t)} \leq \phi_i^{(t-1)}$. Thus $\phi_i^{(t)}$ is nonincreasing and $\phi_i^{\text{fin}} \leq \phi_i^{(0)} = v_{\text{alg}}(\{i\})$. □

**Definition 1** (Algorithmic dummy)**.** *A player $i$ is an* algorithmic dummy *if $i \notin S^{(t)}$ for all iterations $t$.*

**Theorem 3** (Dummy player)**.** *If $i$ is an algorithmic dummy, then*

$$\phi_i^{\text{fin}} \;=\; \phi_i^{(0)} \;=\; v_{\text{alg}}(\{i\}).$$

*Proof.* If $i \notin S^{(t)}$ for all $t$, then $\phi_i^{(t)} = \phi_i^{(t-1)}$ at every iteration. Hence $\phi_i^{\text{fin}} = \phi_i^{(0)} = v_{\text{alg}}(\{i\})$. □

**Remark 1.** *In classical cooperative cost games, a (true) dummy satisfies $v(S \cup \{i\}) = v(S) + v(\{i\})$ for all $S \subseteq M \setminus \{i\}$. If the neighborhoods and data imply that any accepted move involving $i$ and other companies yields zero incremental savings beyond $i$'s stand-alone part, then $i$ is algorithmic dummy and the classical dummy axiom is recovered for the algorithm-induced values.*

**Theorem 4** (Symmetry under strong invariance assumptions)**.** *Let $M$ be the set of companies and let the algorithm described in Algorithms 1–**??** produce the characteristic values $v_{\text{alg}}(\cdot)$ and the iterative allocations $\{\phi^{(t)}\}_{t\geq 0}$. Suppose companies $i, j \in M$ are interchangeable in the sense that*

$$\forall S \subseteq M \setminus \{i, j\} : \quad v_{\text{alg}}(S \cup \{i\}) = v_{\text{alg}}(S \cup \{j\}).$$

*Assume further the following conditions hold for the execution of the algorithm:*

*(A1)* ***Initial allocation symmetry:*** $\phi_i^{(0)} = \phi_j^{(0)}$.

*(A2)* ***Weight symmetry:*** *for every iteration $t$ with affected coalition $S^{(t)}$ such that $i, j \in S^{(t)}$, the cost-sharing weights assigned to $i$ and $j$ are equal:*

$$w_i^{(t)} = w_j^{(t)}.$$

*(A3)* ***Algorithmic invariance / symmetric selection:*** *the local-search procedure (including candidate-generation, evaluation, and any tie-breaking rule) is invariant under the relabelling of $i$ and $j$. Equivalently, whenever a move affecting only $i$ has the same cost reduction as the corresponding relabelled move affecting only $j$, the selection mechanism treats them identically (i.e., it will either accept both simultaneously, choose them with equal probability, or choose neither).*

*Then at termination of the algorithm the final allocations satisfy*

$$\phi_i^{\text{fin}} = \phi_j^{\text{fin}},$$

*i.e. the allocation produced by the algorithm respects the symmetry axiom for the interchangeable players $i$ and $j$.*

*Proof.* We prove $\phi_i^{(t)} = \phi_j^{(t)}$ for all iterations $t \geq 0$ by induction on $t$. The theorem then follows by taking $t =$ the final iteration.

**Base case** ($t = 0$). By assumption (A1) we have $\phi_i^{(0)} = \phi_j^{(0)}$.

**Inductive step.** Assume for some $t \geq 1$ that $\phi_i^{(t-1)} = \phi_j^{(t-1)}$. Consider iteration $t$. Let the accepted move at iteration $t$ (if any) reduce the total cost by $\Delta^{(t)} > 0$ and affect coalition $S^{(t)} \subseteq M$.

We distinguish three exhaustive cases.

*Case 1:* $i \notin S^{(t)}$ and $j \notin S^{(t)}$. Neither allocation changes in iteration $t$, hence

$$\phi_i^{(t)} = \phi_i^{(t-1)} = \phi_j^{(t-1)} = \phi_j^{(t)}.$$

*Case 2:* $i \in S^{(t)}$ and $j \in S^{(t)}$. By the allocation update rule,

$$\phi_i^{(t)} = \phi_i^{(t-1)} - w_i^{(t)}\Delta^{(t)}, \qquad \phi_j^{(t)} = \phi_j^{(t-1)} - w_j^{(t)}\Delta^{(t)}.$$

Given that $\phi_i^{(t-1)} = \phi_j^{(t-1)}$ by the inductive hypothesis and $w_i^{(t)} = w_j^{(t)}$ by the assumption (A2), it follows directly from the update rule that $\phi_i^{(t)} = \phi_j^{(t)}$.

*Case 3:* exactly one of $\{i, j\}$ belongs to $S^{(t)}$. Without loss of generality suppose $i \in S^{(t)}$ and $j \notin S^{(t)}$. Then the allocation update would yield

$$\phi_i^{(t)} = \phi_i^{(t-1)} - w_i^{(t)} \Delta^{(t)}, \qquad \phi_j^{(t)} = \phi_j^{(t-1)}.$$

Thus $\phi_i^{(t)} \neq \phi_j^{(t)}$ in general. We show this case cannot occur under the stated assumptions.

Because $i$ and $j$ are interchangeable (the characteristic function is invariant under swapping $i$ and $j$), the set of candidate moves obtainable from the pooled routes is closed under the relabelling that swaps $i$ and $j$: to every move $m$ there corresponds a relabelled move $m'$ with identical cost reduction, and $m$ affects a set $S$ iff $m'$ affects the relabelled set. By algorithmic invariance (A3), the selection mechanism treats $m$ and $m'$ identically. Therefore the algorithm cannot accept a move that affects $i$ but not $j$ while rejecting the corresponding relabelled move (which affects $j$ but not $i$). Concretely, if a unique best move exists and affects only $i$, by (A3) there must be a symmetric best move affecting only $j$ and the algorithm must either accept both (e.g. by simultaneous application or symmetric tie-breaking) or neither. Hence an accepted move affecting exactly one of $i, j$ is impossible under (A3). Thus Case 3 cannot occur.

Since Case 3 is excluded, only Cases 1 and 2 can happen, and in both cases we have $\phi_i^{(t)} = \phi_j^{(t)}$. This completes the inductive step.

By induction, $\phi_i^{(t)} = \phi_j^{(t)}$ for all $t$, and in particular at termination $\phi_i^{\text{fin}} = \phi_j^{\text{fin}}$. □

## 5.2 Numerical Results

This section presents a comprehensive computational evaluation of the proposed heuristic cost allocation algorithm. The experiments are designed to assess the algorithm's performance along two primary dimensions: (1) the quality and fairness of the resulting cost allocations compared to established cooperative game theory benchmarks, and (2) its computational efficiency and scalability. The analysis is structured into three parts: an evaluation across diverse problem configurations, an investigation of small-scale instances to understand fundamental behaviors, and a scalability analysis to test practical limits. For all experiments, we simplify the parameters by setting each customer's demand to 1 and all vehicle capacities to 3.

### 5.2.1 Analysis of savings and time on small instances

Firstly, to gain insight into the impact of coalition structure, we analyzed small instances with 2 or 3 companies and a limited number of customers. We examined both balanced and heterogeneous customer distributions across companies. Each configuration was solved over 10 independent trials with randomly generated customer locations, and the average savings and the average execution time across these trials are reported. Savings are calculated as $\sum_i \mathcal{R}_i^{\text{fin}}$ The key metrics were the percentage of cost savings achieved through collaboration and the algorithm's execution time.

The results, summarized in Table 1, reveal clear trends. Balanced configurations (e.g., [3,3,3]) consistently yield high savings, approximately 25-26%. Configurations with moderate heterogeneity (e.g., [4,3,5], [5,2,5]) maintain similarly high savings, indicating that collaboration remains beneficial even with some imbalance. Here, a configuration such as [x,y,z] denotes that the first company has $x$ customers, the second company has $y$ customers, and the third company serves $z$ customers.

However, configurations with extreme size disparities, such as [5,1,6] and [7,1,4], show significantly lower savings (15.3% and 17.1%, respectively). This suggests that when one company contributes very few customers to the grand coalition, the potential for synergistic cost sharing diminishes. Interestingly, configuration [7,2,3] achieves high savings (26.1%), indicating that a moderate number of customers from a small partner (2 instead of 1) can restore collaborative benefits, and that geographical factors can outweigh purely size-based disparities. Execution times grow modestly with the total number of customers but remain within practical limits for these small instances.

Table 1: Performance Analysis on Small-Scale Configurations

| **Configuration** | **Companies** | **Customers** | **Avg. Savings (%)** | **Avg. Exec. Time (s)** |
|---|---|---|---|---|
| 1 | 2 | [2, 2] | 20.3 ± 8.8 | 0.09 |
| 2 | 2 | [3, 3] | 13.5 ± 7.6 | 0.29 |
| 3 | 3 | [2, 2, 2] | 21.4 ± 6.6 | 0.35 |
| 4 | 3 | [3, 3, 3] | 25.7 ± 7.2 | 0.78 |
| 5 | 3 | [4, 3, 5] | 23.8 ± 8.2 | 1.84 |
| 6 | 3 | [5, 2, 5] | 23.9 ± 9.5 | 1.98 |
| 7 | 3 | [5, 1, 6] | 15.3 ± 5.4 | 1.67 |
| 8 | 3 | [7, 1, 4] | 17.1 ± 7.4 | 1.75 |
| 9 | 3 | [7, 2, 3] | 26.1 ± 4.4 | 1.65 |

#### 5.2.2 Geographical configuration and comparison with game-theoretic benchmarks

To evaluate the robustness of the proposed framework, we constructed four distinct problem configurations that vary along two critical axes: the geographical distribution of customers (Dense vs. Sparse) and the balance of customers among companies (Homogeneous vs. Heterogeneous). These configurations are defined as follows:

1. **Homogeneous–Dense:** Each company has an equal number of customers, and all customers are located within similar, overlapping regions. A degree of sparsity was introduced to ensure the potential for collaboration.

2. **Homogeneous–Sparse:** Each company has an equal number of customers, but all customers are distributed sparsely across the service area.

3. **Heterogeneous–Dense:** Companies have a different number of customers, but their customers are located within similar, overlapping regions. A degree of sparsity was introduced to ensure collaboration.

4. **Heterogeneous–Sparse:** Companies have a different number of customers, and all customers are distributed sparsely.

Customer locations were generated randomly for each configuration. Figure 2 visualizes the resulting customer distributions, illustrating the distinct spatial structures.

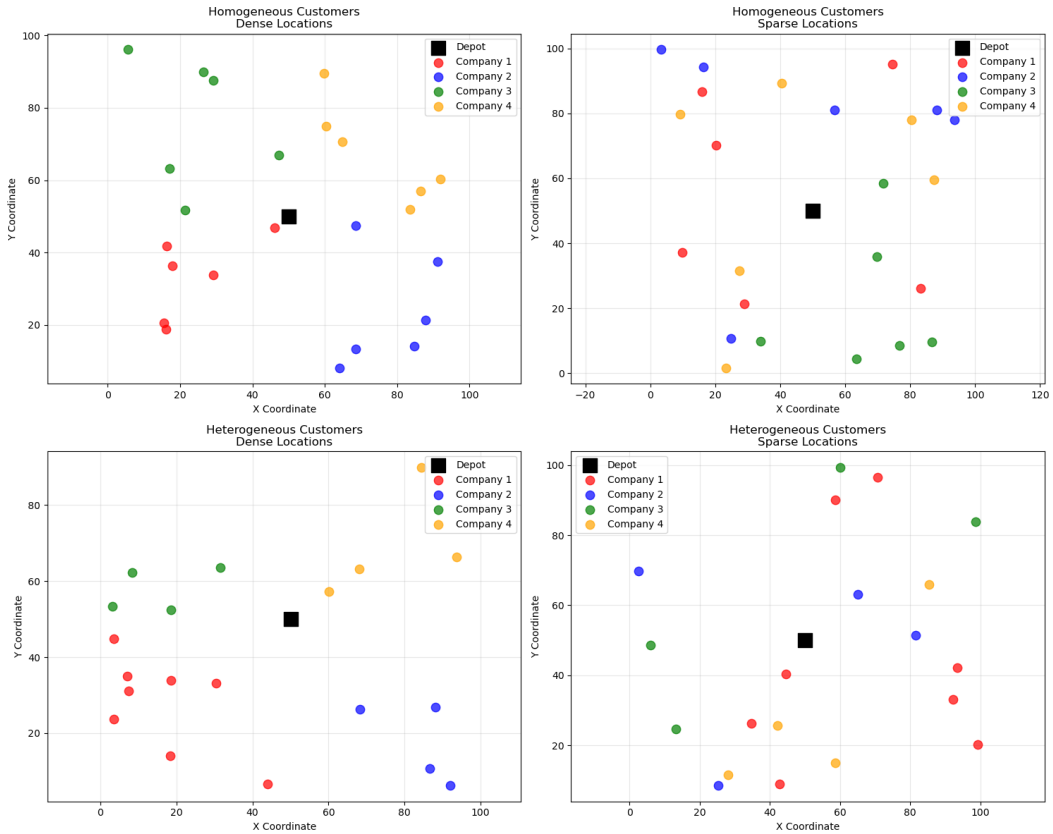


Figure 2: Customer locations for the four problem configurations: Homogeneous-Dense, Homogeneous-Sparse, Heterogeneous-Dense, and Heterogeneous-Sparse.

For each configuration, we computed the cost allocation using our heuristic method and compared it against three classical cooperative game theory solutions: the Shapley value, the Equal Surplus Division (ESD), and the Nucleolus. The results, including the total collaborative cost and the allocation to each company, are presented in Table 2.

Table 2: Cost Allocation Comparison Across Problem Configurations

| Configuration | Total Cost | Co. 1 | Co. 2 | Co. 3 | Co. 4 |
|---|---|---|---|---|---|
| **Homogeneous-Dense** | | | | | |
| Heuristic | 838.64 | 241.59 | 203.26 | 181.30 | 212.49 |
| Shapley | 838.64 | 241.58 | 203.26 | 181.31 | 212.50 |
| ESD | 838.64 | 241.59 | 203.26 | 181.30 | 212.49 |
| Nucleolus | 838.64 | 241.59 | 203.26 | 181.30 | 212.49 |
| **Homogeneous-Sparse** | | | | | |
| Heuristic | 844.71 | 178.13 | 176.62 | 212.86 | 277.09 |
| Shapley | 844.71 | 174.85 | 175.92 | 202.10 | 291.86 |
| ESD | 844.71 | 186.96 | 143.88 | 200.61 | 313.26 |
| Nucleolus | 844.71 | 197.16 | 178.01 | 177.25 | 292.29 |
| **Heterogeneous-Dense** | | | | | |
| Heuristic | 805.59 | 245.25 | 164.47 | 167.50 | 228.38 |
| Shapley | 805.59 | 240.23 | 149.13 | 154.09 | 211.86 |
| ESD | 805.59 | 232.68 | 151.90 | 154.93 | 215.81 |
| Nucleolus | 805.59 | 245.25 | 140.74 | 164.68 | 204.65 |
| **Heterogeneous-Sparse** | | | | | |
| Heuristic | 779.86 | 348.81 | 165.43 | 173.83 | 91.79 |
| Shapley | 779.86 | 328.74 | 142.15 | 156.63 | 106.14 |
| ESD | 779.86 | 347.99 | 131.01 | 155.82 | 98.83 |
| Nucleolus | 779.86 | 329.76 | 137.60 | 144.31 | 121.98 |

The results demonstrate the algorithm's sensitivity to problem structure. In the **Homogeneous–Dense** configuration, the heuristic produces an allocation nearly identical to the Nucleolus and very close to the Shapley value. This alignment with core game-theoretic concepts under balanced, dense conditions indicates the heuristic's ability to capture fair cost-sharing when collaboration synergies are low and symmetrical.

In the **Homogeneous–Sparse** configuration, the total cost is higher due to increased travel distances. The allocations across all methods show greater variation, reflecting the geographical disparities. The heuristic's allocation for Company 4 is notably high, suggesting it bears a larger share of the burden imposed by its customers' remote locations, a trend also observed, though to a different extent, in the benchmark methods.

For the **Heterogeneous–Dense** case, the heuristic's results are closest to the Shapley value. The proximity of Companies 1 and 3 (and similarly, Companies 2 and 4) creates natural coalition incentives, which both the heuristic and Shapley value appear to capture effectively. In contrast, the Nucleolus produces a more distinct allocation, potentially prioritizing different stability criteria (coalitional rationality).

Unsurprisingly, the **Heterogeneous–Sparse** configuration yields the high overall cost, as the combination of imbalanced company sizes and dispersed customers maximizes routing complexity. The heuristic allocation is once again close to the Shapley value,

demonstrating consistent behavior across heterogeneous settings.

Table 3: Individual Savings from Cooperation under the Heuristic Allocation (%)

| **Configuration** | **Co. 1** | **Co. 2** | **Co. 3** | **Co. 4** |
|---|---|---|---|---|
| Homogeneous-Dense | 0.00 | 0.00 | 0.00 | 0.00 |
| Homogeneous-Sparse | 25.66 | 10.13 | 15.95 | 24.27 |
| Heterogeneous-Dense | 0.00 | 0.00 | 0.00 | 0.00 |
| Heterogeneous-Sparse | 8.79 | 0.00 | 8.62 | 31.11 |

Table 3 reports the individual percentage savings achieved by each company under the proposed heuristic allocation, relative to their non-cooperative costs. The results highlight that the benefits of cooperation are highly configuration-dependent. In dense instances, both homogeneous and heterogeneous, savings are essentially negligible for all companies, indicating that collaboration does not substantially improve cost efficiency in already well-structured networks. In contrast, sparse configurations exhibit significant gains from cooperation, particularly under homogeneous demand distributions, where all companies experience double digit savings. The heterogeneous sparse case shows a more uneven distribution of benefits: while some companies obtain substantial reductions in cost, others experience marginal or near zero gains. These findings suggest that the heuristic allocation captures meaningful cooperative synergies when network sparsity limits individual routing efficiency, while maintaining near budget-balance in scenarios with limited collaboration potential.

#### 5.2.3 Scalability analysis

We evaluated the algorithm's performance across different problem sizes to assess its practical applicability and confirm our central claim: that the heuristic allocation yields significantly more efficient solutions compared to established cooperative game theory concepts, specifically the Shapley value and the nucleolus.

Our initial experiment analyzed how computational time changes as the total number of customer sets increases. We maintained a fixed setup of three cooperating companies (all customers distributed among them), fixed the vehicle capacity at 5, and simplified customer demand to 1 for all customers. We tested instance sizes ranging from 100 to 1000 total customers, using the average result from 5 randomly generated instances for each size.

As Figure 3 illustrates, the computational time for all allocations increases exponentially, which is expected since solving the VRP is NP-hard. An interesting finding is the extremely close computational time between the Shapley value and the nucleolus across all instances. This suggests that the primary computational bottleneck for both is the calculation of the characteristic function; the time consumed by this step hardly dominates all other calculations, such as sequential Linear Programs and permutations.

Nevertheless, the heuristic allocation remains consistently and significantly more efficient. As the customer set size grows, the difference in computational time increasingly favors the heuristic. Notably, when the total number of customers reaches 1000, the heuristic allocation is almost 50% faster than both the nucleolus and the Shapley value.

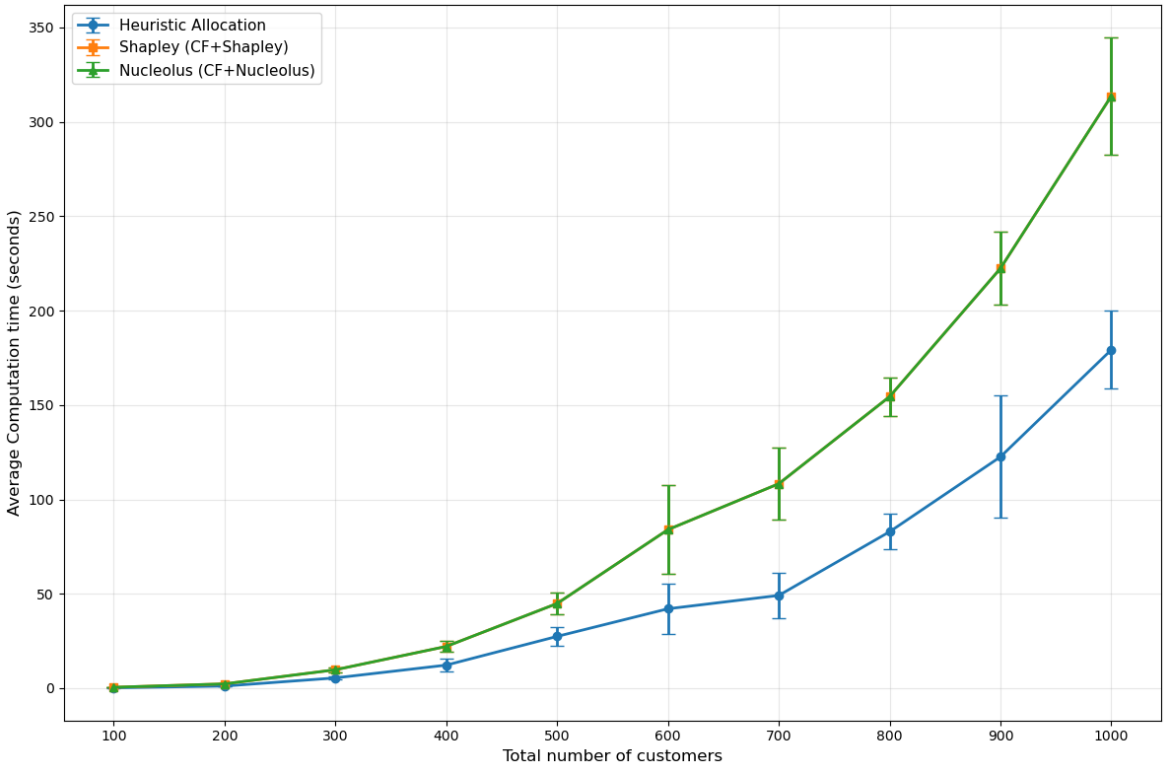


Figure 3: Comparison of the average computation time (seconds) for the heuristic allocation, Shapley value, and nucleolus as the total number of customers increases.

It's important to note that the computational bottleneck in the characteristic function calculation stems from the increase in the number of companies (consequently, number of coalitions) rather than the number of customers. Therefore, we conducted a second experiment to specifically demonstrate the computational time of all allocations across a varying number of companies.

In this experiment, we fixed the number of customers at 100 and distributed them among the cooperating companies. We used the same set of customer locations across all company configurations (from 2 companies to 10 companies) and reported the average computational time from 5 different customer location setups. As shown in Figure 4, the heuristic allocation scales linearly as the number of companies increases, while the other two solution concepts escalate exponentially. The line representing the heuristic allocation is very close to zero because this experiment was conducted only for 100 customers, which was chosen to simplify the computation for the other methods. These results strongly confirm the superior scalability of the heuristic, especially in coalition

formation settings involving a large number of partners.

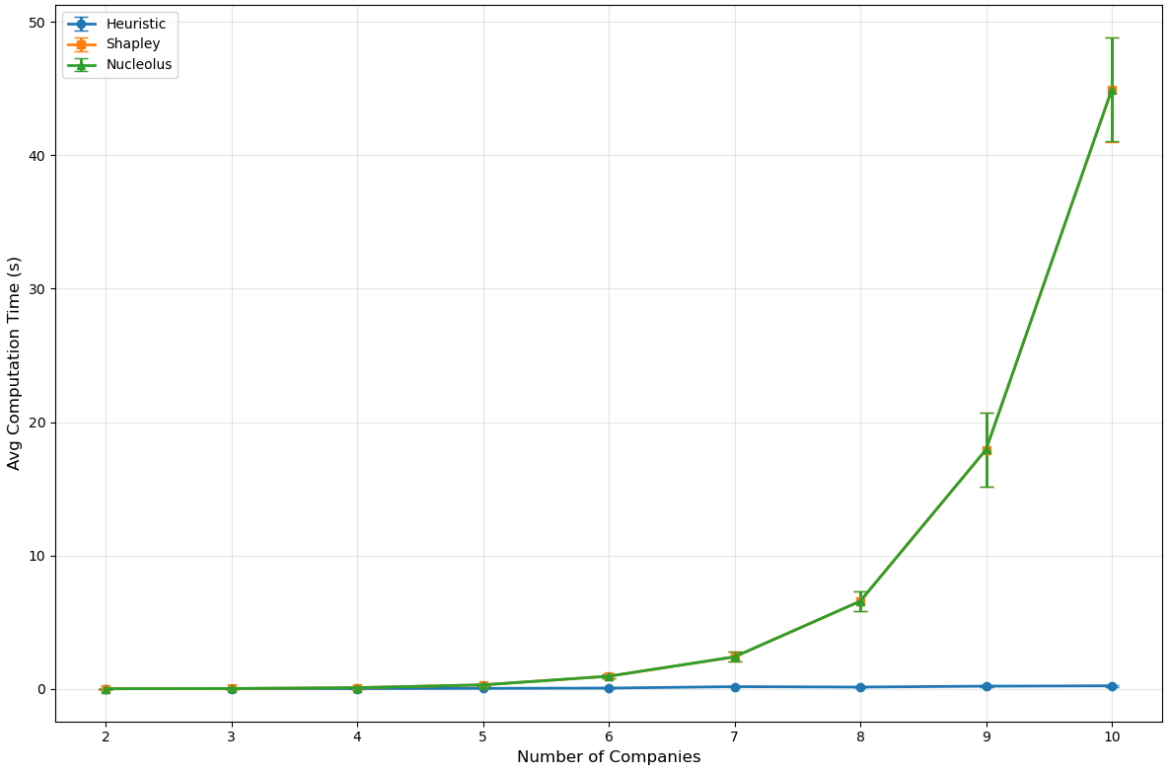


Figure 4: Comparison of the average computation time (seconds) for the heuristic allocation, Shapley value, and nucleolus as the total number of companies increases.

# 6 Conclusion

This paper introduced a novel heuristic framework for simultaneously solving the collaborative Vehicle Routing Problem (VRP) and allocating costs fairly among participating companies. Unlike classical cooperative game theoretic approaches, our method does not rely on the explicit computation of the characteristic function, which is computationally prohibitive for large instances. Instead, it combines Clarke–Wright savings for initial route construction with a deterministic local search improvement phase, where cost savings are dynamically distributed among companies through a proportional allocation rule.

From a theoretical perspective, we established that the proposed algorithm satisfies key cooperative game theoretic properties such as efficiency, individual rationality, and the dummy player axiom, and respects symmetry under certain invariance conditions. These properties provide a strong fairness foundation for the allocation mechanism.

The numerical experiments on small instances demonstrated that our heuristic produces cost allocations that are very close to classical benchmarks such as the Shapley

value, the Nucleolus, and Equal Surplus Division, particularly in balanced and dense configurations. Furthermore, the algorithm achieves substantial cost savings with reasonable computation times, making it well-suited for practical applications where exact methods are infeasible. The scalability analysis further highlighted its potential to handle larger problem instances while maintaining meaningful economic savings.

Future research may focus on several extensions. First, applying the heuristic to real-world collaborative logistics data would provide valuable insights into its practical performance and acceptance. Second, further refinement of the stochastic variant could enhance solution quality for large-scale problems. Finally, integrating negotiation or privacy-preserving mechanisms would make the method applicable to decentralized and information-sensitive collaboration settings.

Overall, the proposed approach offers a fair and computationally efficient alternative to traditional cooperative game theoretic methods, bridging the gap between theoretical allocation principles and practical collaborative vehicle routing.

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